

Chapitre 7 : La comparaison de deux groupes

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1 Avant de commencer

Ce chapitre reprends les codes du chapitre 7 sur *la comparaison de deux groupes* du livre **Langage R et Statistiques : Initiation à l'analyse de données** publié par les éditions ENI (2^{ème} édition).

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.0      ✓ readr      2.1.6
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2     4.0.2      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr       1.2.1
— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

2 Comparaison de proportion

2.1 Test paramétrique

```
prop.test(x = c(42, 68), n = c(139, 175))
```

```
2-sample test for equality of proportions with continuity correction

data:  c(42, 68) out of c(139, 175)
X-squared = 2.1762, df = 1, p-value = 0.1402
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.19795128  0.02512497
sample estimates:
   prop 1    prop 2 
0.3021583 0.3885714
```

```
prop.test(x = c(42, 68, 24), n = c(139, 175, 135))
```

```

3-sample test for equality of proportions without continuity correction

data:  c(42, 68, 24) out of c(139, 175, 135)
X-squared = 16.187, df = 2, p-value = 0.0003056
alternative hypothesis: two.sided
sample estimates:
  prop 1    prop 2    prop 3 
0.3021583 0.3885714 0.1777778

```

2.2 Test exact de Fisher

```

(gendre_ratio_startup <-
  matrix(
    c(1, 2, 5, 7),
    nrow = 2,
    dimnames =
      list(
        c("Comite", "Salaries"),
        c("Femme", "Homme")
      )
  ))

```

	Femme	Homme
Comite	1	5
Salaries	2	7

```
fisher.test(gendre_ratio_startup)
```

```

Fisher's Exact Test for Count Data

data:  gendre_ratio_startup
p-value = 1
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.009917539 17.649421539
sample estimates:
odds ratio
 0.716429

```

reprise de l'exemple de la page d'aide de `fisher.test()` : table Agresti (2002, p. 57) Job Satisfaction

```
(Job <- matrix(
  c(1,2,1,0, 3,3,6,1, 10,10,14,9, 6,7,12,11),
  4,
  4,
  dimnames =
    list(
      salaire = c("< 15k", "15-25k", "25-40k", "> 40k"),
      satisfaction = c("tres decevant", "un peu decevant", "moderement satisfait",
        "très satisfait")
    )
))
```

	satisfaction			
salaire	tres decevant	un peu decevant	moderement satisfait	très satisfait
< 15k	1	3	10	6
15-25k	2	3	10	7
25-40k	1	6	14	12
> 40k	0	1	9	11

```
fisher.test(Job)
```

Fisher's Exact Test for Count Data

```
data: Job
p-value = 0.7827
alternative hypothesis: two.sided
```

2.3 Test de Mac Nemar

```
(test_allergene <- matrix(
  c(23, 14, 84, 59),
  nrow = 2,
  dimnames = list(avant_traitement = c("réaction allergique", "réaction non
    allergique"),
                  après_traitement = c("réaction allergique", "réaction
    non allergique")
  )
))
```

avant_traitement	après_traitement	
	réaction allergique	réaction non allergique
réaction allergique	23	84
réaction non allergique	14	59

```
mcnemar.test(test_allergene)
```

McNemar's Chi-squared test with continuity correction

data: test_allergene

McNemar's chi-squared = 48.582, df = 1, p-value = 3.168e-12

```
prop.test(test_allergene)
```

2-sample test for equality of proportions with continuity correction

data: test_allergene

X-squared = 0.036067, df = 1, p-value = 0.8494

alternative hypothesis: two.sided

95 percent confidence interval:

-0.1075760 0.1539209

sample estimates:

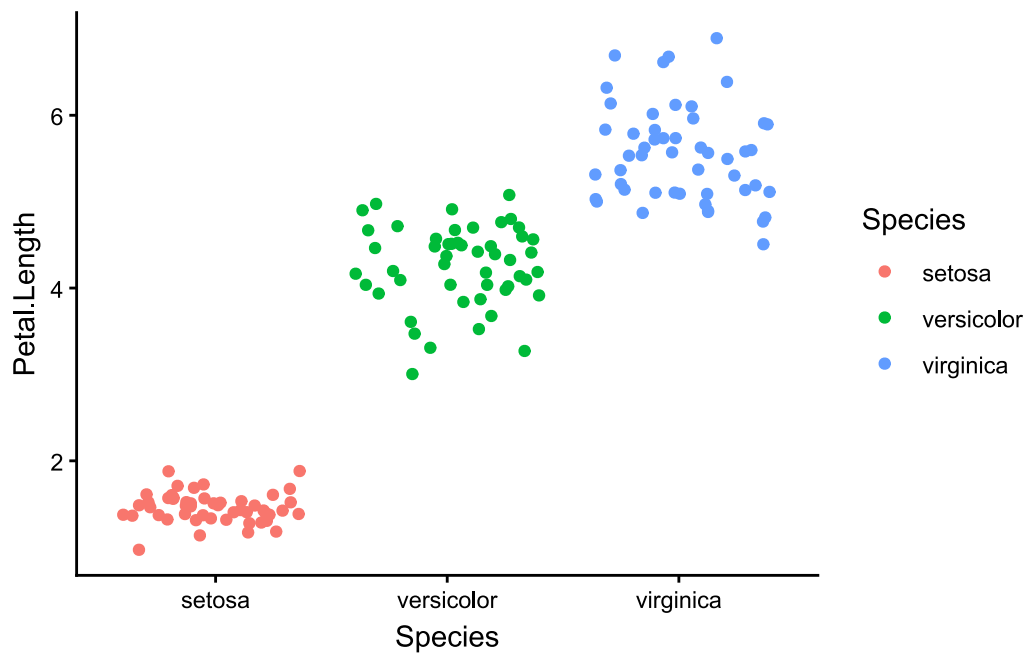
prop 1 prop 2

0.2149533 0.1917808

3 Comparaison de variances

3.1 graphique initial

```
ggplot(iris) +
  aes(x = Species, y = Petal.Length, color = Species) +
  geom_jitter() +
  theme_classic()
```



3.2 Test F

vérification de la normalité

```
filter(mtcars, am == 0)$mpg |> shapiro.test()
```

Shapiro-Wilk normality test

```
data: filter(mtcars, am == 0)$mpg
W = 0.97677, p-value = 0.8987
```

```
filter(mtcars, am == 1)$mpg |> shapiro.test()
```

Shapiro-Wilk normality test

```
data: filter(mtcars, am == 1)$mpg
W = 0.9458, p-value = 0.5363
```

test

```
var.test(filter(mtcars, am == 0)$mpg, filter(mtcars, am == 1)$mpg)
```

F test to compare two variances

```
data: filter(mtcars, am == 0)$mpg and filter(mtcars, am == 1)$mpg
F = 0.38656, num df = 18, denom df = 12, p-value = 0.06691
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1243721 1.0703429
sample estimates:
ratio of variances
 0.3865615
```

3.3 Test de Bartlett

iris

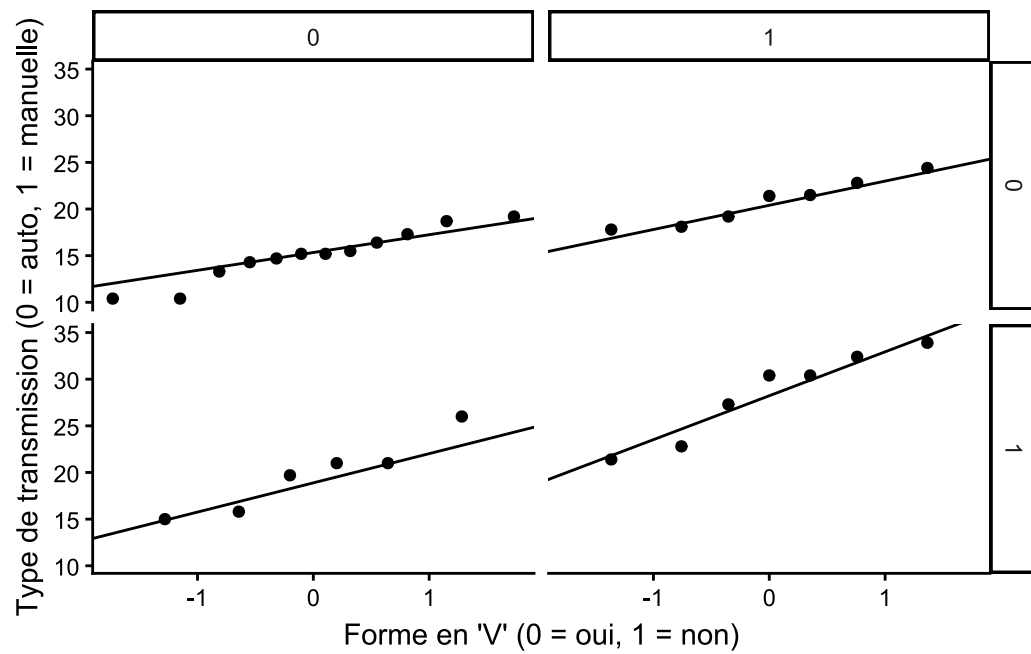
```
bartlett.test(Petal.Length ~ Species, data = iris)
```

Bartlett test of homogeneity of variances

```
data: Petal.Length by Species
Bartlett's K-squared = 55.423, df = 2, p-value = 9.229e-13
```

mtcars : vérification de la normalité

```
ggplot(mtcars) +
  aes(sample = mpg) +
  geom_qq() +
  geom_qq_line() +
  facet_grid(rows = vars(am), cols = vars(vs)) +
  labs(x = "Forme en 'V' (0 = oui, 1 = non)", y = "Type de transmission (0 =
auto, 1 = manuelle)") +
  theme_classic()
```



mtcars : test de bartlett

```
bartlett.test(mpg ~ interaction(am, vs), data = mtcars)
```

Bartlett test of homogeneity of variances

data: mpg by interaction(am, vs)

Bartlett's K-squared = 3.5598, df = 3, p-value = 0.3131

3.4 Test de Levene

vérification de la normalité au sein de chaque espèce

```
filter(iris, Species == "setosa")$Petal.Length |> shapiro.test()
```

Shapiro-Wilk normality test

data: filter(iris, Species == "setosa")\$Petal.Length

W = 0.95498, p-value = 0.05481

```
filter(iris, Species == "versicolor")$Petal.Length |> shapiro.test()
```

Shapiro-Wilk normality test

```
data: filter(iris, Species == "versicolor")$Petal.Length  
W = 0.966, p-value = 0.1585
```

```
filter(iris, Species == "virginica")$Petal.Length |> shapiro.test()
```

Shapiro-Wilk normality test

```
data: filter(iris, Species == "virginica")$Petal.Length  
W = 0.96219, p-value = 0.1098
```

```
library(car)
```

Le chargement a nécessité le package : carData

Attachement du package : 'car'

L'objet suivant est masqué depuis 'package:dplyr':

recode

L'objet suivant est masqué depuis 'package:purrr':

some

```
leveneTest(Petal.Length ~ Species, data = iris)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	2	19.48	3.129e-08 ***
	147		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

3.5 Test de Fligner-Killeen

vérification de la normalité des données pour chaque spray

```
purrr::map(
  .x = LETTERS[1:6],
  .f = ~ filter(InsectSprays, spray == .x)$count |> shapiro.test()
) |>
  set_names(glue::glue("Insecticide {LETTERS[1:6]}"))
```

```
$`Insecticide A`
```

```
Shapiro-Wilk normality test
```

```
data: filter(InsectSprays, spray == .x)$count
W = 0.95757, p-value = 0.7487
```

```
$`Insecticide B`
```

```
Shapiro-Wilk normality test
```

```
data: filter(InsectSprays, spray == .x)$count
W = 0.95031, p-value = 0.6415
```

```
$`Insecticide C`
```

```
Shapiro-Wilk normality test
```

```
data: filter(InsectSprays, spray == .x)$count
W = 0.85907, p-value = 0.04759
```

```
$`Insecticide D`
```

```
Shapiro-Wilk normality test
```

```
data: filter(InsectSprays, spray == .x)$count
W = 0.75063, p-value = 0.002713
```

```
$`Insecticide E`
```

Shapiro-Wilk normality test

```
data: filter(InsectSprays, spray == .x)$count  
W = 0.92128, p-value = 0.2967
```

\$`Insecticide F`

Shapiro-Wilk normality test

```
data: filter(InsectSprays, spray == .x)$count  
W = 0.88475, p-value = 0.1009
```

test

```
fligner.test(count ~ spray, data = InsectSprays)
```

Fligner-Killeen test of homogeneity of variances

```
data: count by spray  
Fligner-Killeen:med chi-squared = 14.483, df = 5, p-value = 0.01282
```

4 Comparaison de moyennes

4.1 Test de Student

```
t.test(filter(mtcars, am == 0)$mpg, filter(mtcars, am == 1)$mpg, var.equal =  
TRUE)
```

Two Sample t-test

```
data: filter(mtcars, am == 0)$mpg and filter(mtcars, am == 1)$mpg  
t = -4.1061, df = 30, p-value = 0.000285  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
-10.84837 -3.64151  
sample estimates:  
mean of x mean of y  
17.14737 24.39231
```

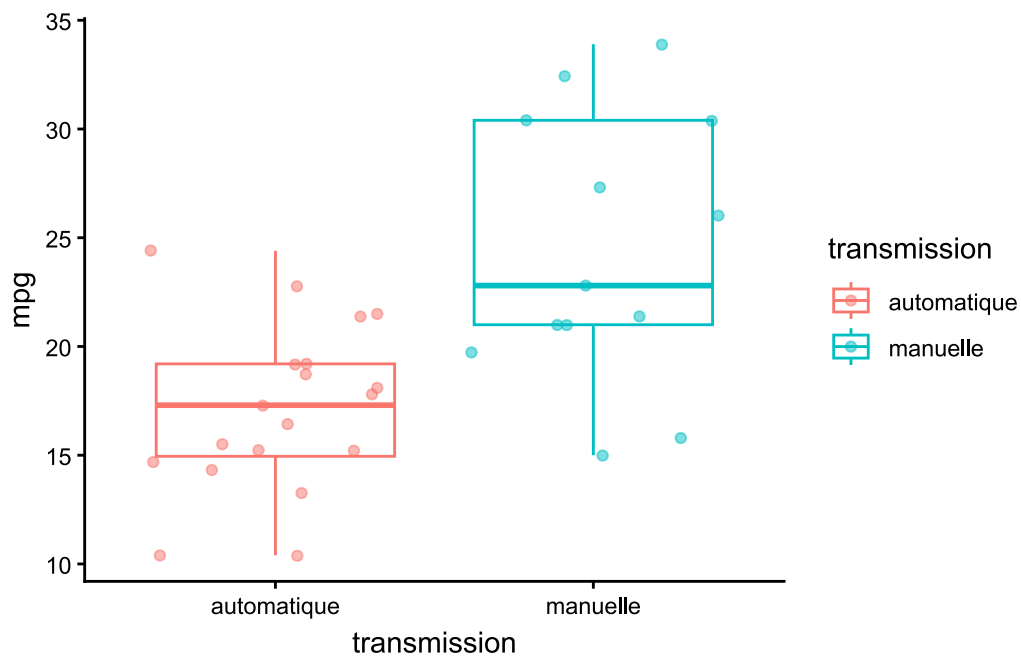
```
t.test(filter(mtcars, am == 0)$mpg, filter(mtcars, am == 1)$mpg, alternative =
"less", var.equal = TRUE)
```

Two Sample t-test

```
data: filter(mtcars, am == 0)$mpg and filter(mtcars, am == 1)$mpg
t = -4.1061, df = 30, p-value = 0.0001425
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
    -Inf -4.250255
sample estimates:
mean of x mean of y
 17.14737  24.39231
```

graph

```
mtcars |>
  mutate(
    transmission =
      case_when(
        am == 0 ~ "automatique",
        am == 1 ~ "manuelle"
      )
  ) |>
  ggplot() +
  aes(x = transmission, y = mpg, color = transmission) +
  geom_boxplot() +
  geom_jitter(alpha = 0.5) +
  theme_classic()
```



4.2 Test de Welch

```
t.test(Petal.Length ~ Species, data = iris |> filter(Species != "virginica"))
```

Welch Two Sample t-test

data: Petal.Length by Species

t = -39.493, df = 62.14, p-value < 2.2e-16

alternative hypothesis: true difference in means between group setosa and group versicolor is not equal to 0

95 percent confidence interval:

-2.939618 -2.656382

sample estimates:

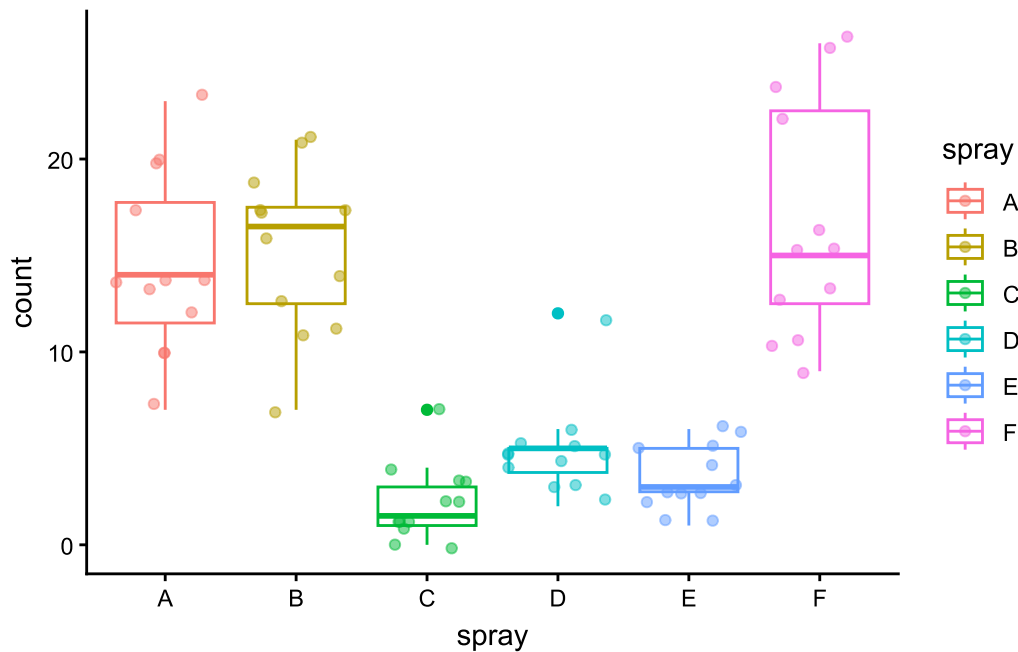
mean in group setosa	mean in group versicolor
1.462	4.260

4.3 Test de Wilcoxon

graph

```
InsectSprays |>
  ggplot() +
  aes(x = spray, y = count, color = spray) +
```

```
geom_boxplot() +
geom_jitter(alpha = 0.5) +
theme_classic()
```



test de wilcoxon

```
wilcox.test(count ~ spray, data = InsectSprays |> filter(spray %in% c("E", "F")))
```

Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): impossible de calculer la p-value exacte avec des ex-aequos

Wilcoxon rank sum test with continuity correction

data: count by spray

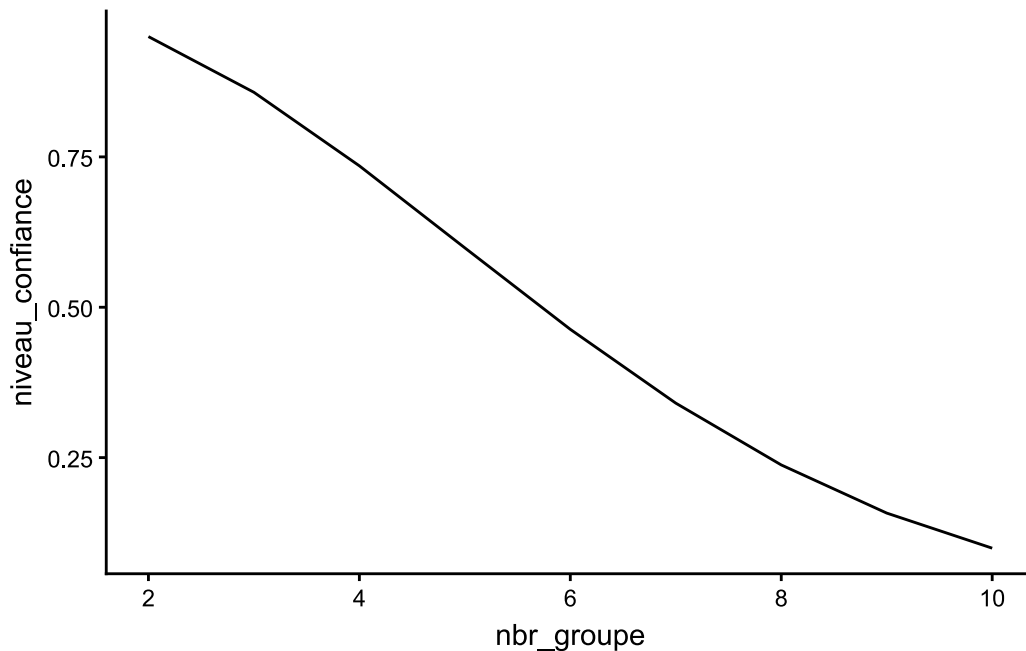
W = 0, p-value = 3.436e-05

alternative hypothesis: true location shift is not equal to 0

5 Comparaison de moyenne de plus de deux groupes

5.1 Erreurs quand le nombre de test est multiplié

```
tibble(  
  nbr_groupe = 2:10,  
  niveau_confiance = (0.95)^choose(nbr_groupe, 2)  
) |>  
  ggplot() +  
  aes(x = nbr_groupe, y = niveau_confiance) +  
  geom_line() +  
  theme_classic()
```



5.2 ANOVA

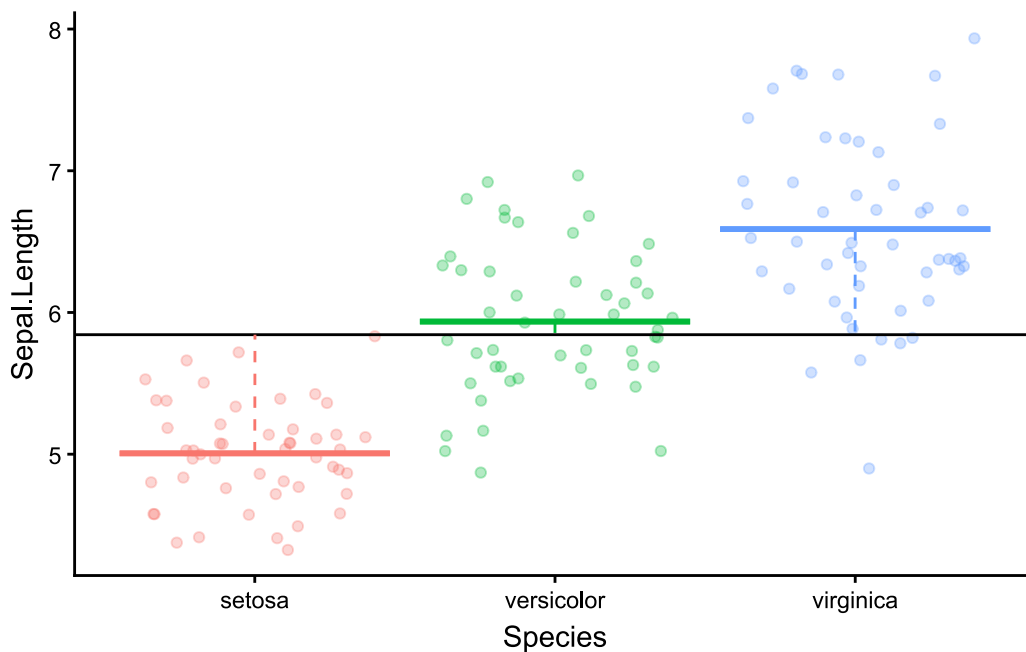
graph explication ANOVA

```
iris_moyenne <- iris |>  
  group_by(Species) |>  
  summarise(moyenne = mean(Sepal.Length))  
  
ggplot(iris) +  
  aes(x = Species, y = Sepal.Length, color = Species) +  
  geom_jitter(alpha = 0.3) +  
  geom_hline(aes(yintercept = mean(Sepal.Length))) +
```

```

    geom_errorbar(aes(ymin = moyenne, y = 5.84, ymax = moyenne), data =
iris_moyenne, linewidth = 1) +
    geom_spoke(aes(y = moyenne, radius = mean(iris$Sepal.Length) - moyenne, angle
= 1.57), data = iris_moyenne, linetype = "dashed") +
    theme_classic() +
    theme(legend.position = "none")

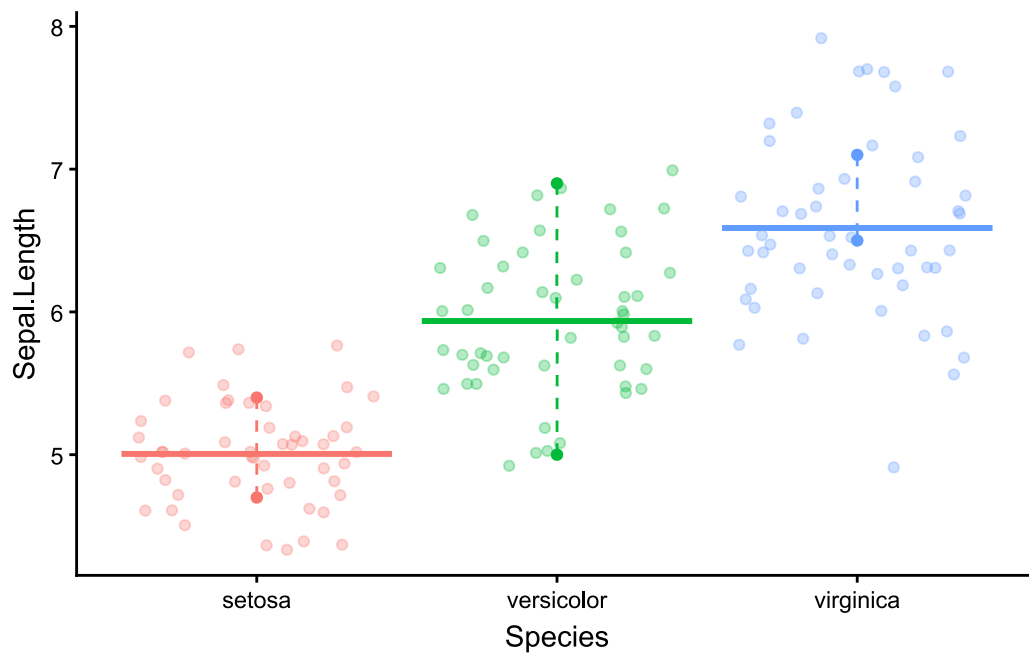
```



```

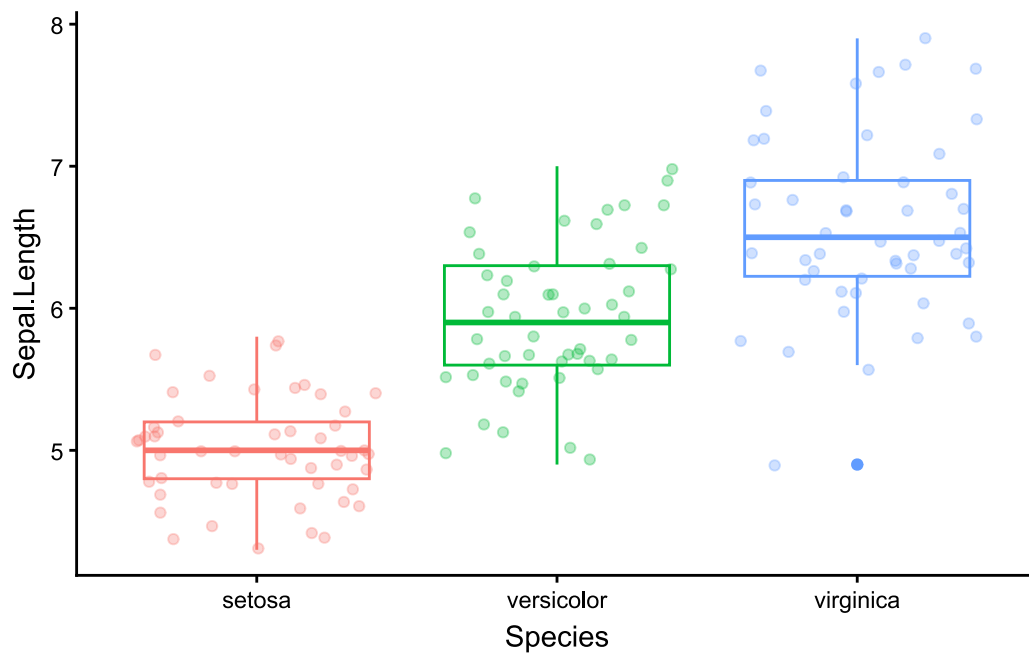
ggplot(iris) +
  aes(x = Species, y = Sepal.Length, color = Species) +
  geom_jitter(alpha = 0.3) +
  geom_errorbar(aes(ymin = moyenne, y = 5.84, ymax = moyenne), data =
iris_moyenne, linewidth = 1) +
  geom_spoke(aes(radius = iris_moyenne$moyenne - Sepal.Length, angle = 1.57),
data = iris |> group_by(Species) |> slice(3) |> ungroup(), linetype = "dashed") +
  geom_spoke(aes(radius = iris_moyenne$moyenne - Sepal.Length, angle = 1.57),
data = iris |> group_by(Species) |> slice(11) |> ungroup(), linetype = "dashed") +
  geom_point(data = iris |> group_by(Species) |> slice(c(3, 11)) |> ungroup()) +
  theme_classic() +
  theme(legend.position = "none")

```



vérification de la normalité et de l'homogénéité des variances

```
ggplot(iris) +  
  aes(x = Species, y = Sepal.Length, color = Species) +  
  geom_boxplot() +  
  geom_jitter(alpha = 0.3) +  
  theme_classic() +  
  theme(legend.position = "none")
```



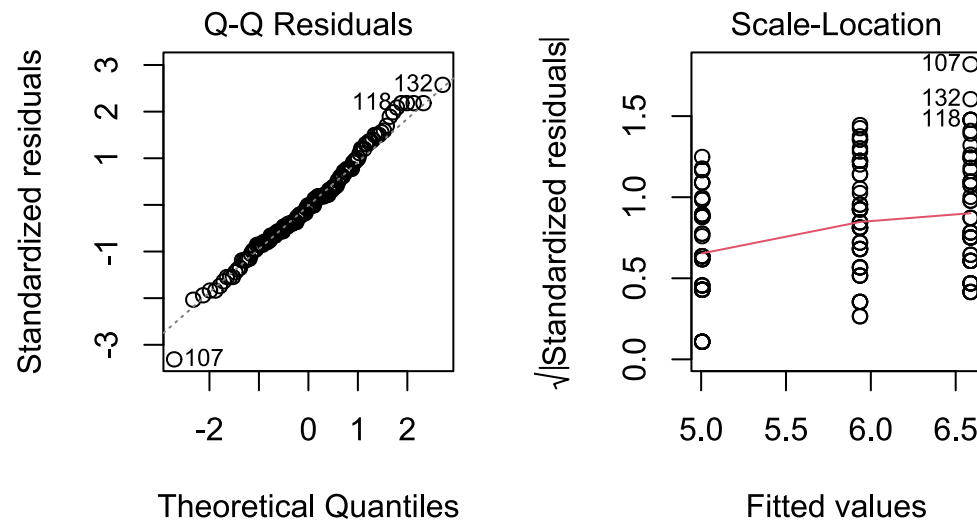
réalisation de l'ANOVA

```
iris_sepal_length_anova <- aov(Sepal.Length ~ Species, data = iris)
summary(iris_sepal_length_anova)
```

```
      Df Sum Sq Mean Sq F value Pr(>F)
Species    2  63.21   31.606   119.3 <2e-16 ***
Residuals 147   38.96    0.265
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

validation du test réalisé

```
par(mfrow=c(1,2))
plot(iris_sepal_length_anova, 2)
plot(iris_sepal_length_anova, 3)
```



post-hoc

```
rstatix::tukey_hsd(iris_sepal_length_anova)
```

```
# A tibble: 3 × 9
  term      group1      group2 null.value estimate conf.low conf.high  p.adj
* <chr>   <chr>      <chr>      <dbl>    <dbl>    <dbl>    <dbl>  <dbl>
1 Species setosa    versicolor      0      0.93     0.686     1.17 3.39e-14
2 Species setosa    virginica       0      1.58     1.34     1.83 3 e-15
3 Species versicolor virginica       0      0.652     0.408     0.896 8.29e- 9
# i 1 more variable: p.adj.signif <chr>
```

5.3 Test de Kruskal-Wallis

```
kruskal.test(count ~ spray, data = InsectSprays)
```

Kruskal-Wallis rank sum test

data: count by spray

Kruskal-Wallis chi-squared = 54.691, df = 5, p-value = 1.511e-10

Post-hoc de Nemenyi

```
PMCMRplus::kwAllPairsNemenyiTest(count ~ spray, data = InsectSprays)
```

```
Warning in kwAllPairsNemenyiTest.default(c(10, 7, 20, 14, 14, 12, 10, 23, :  
Ties are present, p-values are not corrected.
```

Pairwise comparisons using Tukey-Kramer-Nemenyi all-pairs test with Tukey-Dist approximation

```
data: count by spray
```

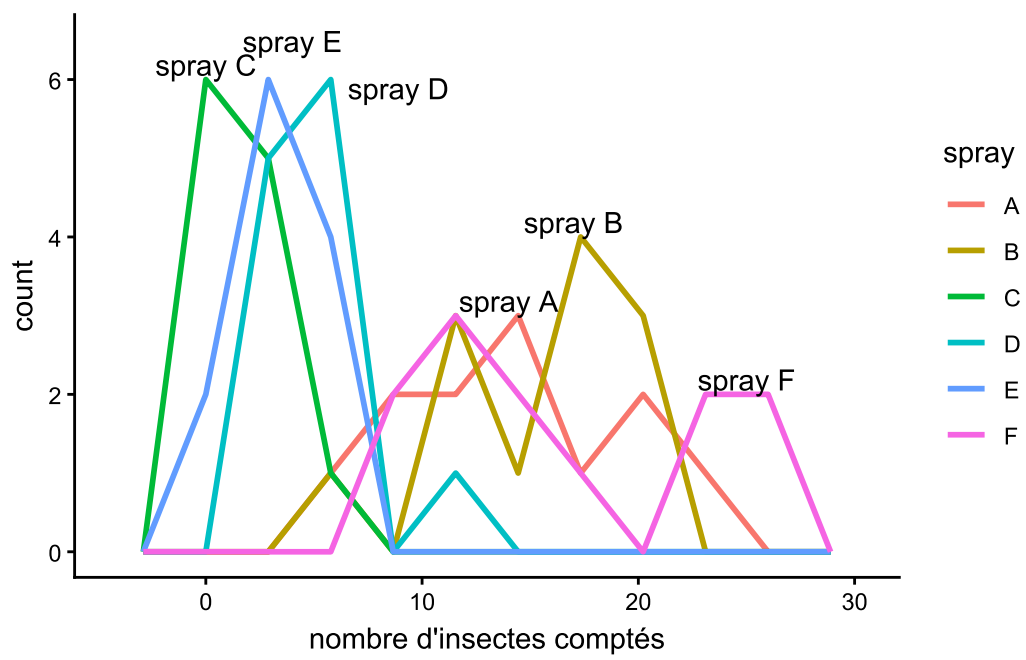
	A	B	C	D	E
B	0.99961	-	-	-	-
C	2.8e-05	5.7e-06	-	-	-
D	0.02293	0.00813	0.56300	-	-
E	0.00169	0.00047	0.94109	0.97809	-
F	0.99861	1.00000	3.5e-06	0.00585	0.00031

```
P value adjustment method: single-step
```

```
alternative hypothesis: two.sided
```

graph

```
ggplot(InsectSprays) +  
  aes(color = spray, x = count) +  
  geom_freqpoly(bins = 10, linewidth = 1) +  
  labs(x = "nombre d'insectes comptés") +  
  annotate("text", x = 0, y = 6.2, label = "spray C") +  
  annotate("text", x = 4, y = 6.5, label = "spray E") +  
  annotate("text", x = 8.9, y = 5.9, label = "spray D") +  
  annotate("text", x = 17, y = 4.2, label = "spray B") +  
  annotate("text", x = 14, y = 3.2, label = "spray A") +  
  annotate("text", x = 25, y = 2.2, label = "spray F") +  
  theme_classic()
```



```
ggsave("img/07Rl32N.png", width = 14, height = 10, units = "cm")
```